MA378: Tutorial Sheet 2 ANS with some solutions

In all the questions below, when needed, the function f is
$$f(x) = \ln(x)$$

1. (a) Write down the Lagrange form for the polynomial, $p_2(x)$, that interpolates f at the points $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$.

Answer: The Lagrange form of the interpolant of degree 2 to f is

$$p_2(x) = f(x_0)L_0(x) + f(x_1)L_1(x) + f(x_2)L_2(x).$$

For this problem

$$L_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2)(x-3)}{(1-2)(1-3)} = \frac{1}{2}(x-2)(x-3),$$

$$L_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-3)}{(2-1)(2-3)} = -(x-1)(x-3),$$

$$L_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x-2)}{(3-1)(3-2)} = \frac{1}{2}(x-1)(x-2).$$

Using that $f(x_0) = f(1) = \ln(1) = 0$, $f(x_1) = f(2) = \ln(2)$ and $f(x_2) = f(3) = \ln(3)$, we get that the Lagrange form is

$$p_2(x) = -\ln(2)(x-1)(x-3) + \ln(3)\frac{(x-1)(x-2)}{2}$$

Notes: It is not required to simplify p_2 any further. In fact, you would get full marks for writing the solution as $p_2(x) = \ln(1)L_0(x) + \ln(2)L_1(x) + \ln(3)L_2(x)$, providing the formulae for the L_i are given.

(b) What bound does Equation (1) give for $|f(4/3) - p_2(4/3)|$?

Answer: Equation (1) gives $f(4/3) - p_2(4/3) = \dfrac{f'''(\tau)}{3!}\pi_3(4/3)$, for some (unknown) $\tau \in (1, 3)$. So the bound on $|f(4/3) - p_2(4/3)|$ is

$$|f(4/3) - p_2(4/3)| \le \frac{\max_{1 \le x \le 3} |f'''(x)|}{3!} |\pi_3(4/3)|. \tag{*}$$

Differentiating f, we get $f'''(x) = 2/x^3$. Since this is a positive function that is decreasing on $[1, 3]$, we get

$$\max_{1 \le x \le 3} |f'''(x)| = f'''(1) = 2/(1)^3 = 2.$$

Also, $1/(3!) = 1/6$. And $\pi_3(x) = (x-1)(x-2)(x-3)$, so $\pi_3(4/3) = (1/3)(-2/3)(-5/3) = 10/27$. Substituting into (*), gives

$$|f(4/3) - p_2(4/3)| \le \left(\frac{2}{6}\right)\left(\frac{10}{27}\right) = \frac{10}{81} \simeq 0.1234.$$

. . . .

2. (a) Give a formula for the piecewise linear interpolant, $l(x)$, that interpolates f, at the points $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$.

Answer:

$$l(x) = \begin{cases} l_1(x) & x_0 \leq x \leq x_1 \\ l_2(x) & x_1 < x \leq x_2 \\ 0 & \text{otherwise} \end{cases}$$

where

$$l_i(x) = f(x_{i-1})\frac{x_i - x}{h} + f(x_i)\frac{x - x_{i-1}}{h},$$

and $h = x_i - x_{i-1}$.

Using that $f(x_0) = 0$, $f(x_1) = \ln(2)$, $f(x_2) = \ln(3)$, and $h = 1$, we get that this simplifies as

$$l(x) = \begin{cases} \ln(2)(x-1) & 1 \leq x \leq 2 \\ \ln(2)(3-x) + \ln(3)(x-2) & 2 < x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

(b) Use Equation (2) to give an upper bound for $\|f(x) - l(x)\|_\infty$.

Answer: (2) is

$$\|f - l\|_\infty \leq \frac{h^2}{8}\|f''\|_\infty.$$

In this instance $h = 1$. Also, $f''(x) = -1/x^2$, so $\|f''\|_\infty = |f''(1)| = 1/(1)^2 = 1$. So

$$\|f - l\|_\infty \leq \frac{(1)^2}{8}(1) = \frac{1}{8} = 0.125.$$

(c) What value of N would you have to choose so that $\|f - l\|_\infty \leq 10^{-6}$?

Answer: We need h such that $\frac{h^2}{8}\|f''\|_\infty \leq 10^{-6}$. That is $h \leq \sqrt{8 \times 10^{-6}} = 0.002828$. Also, $h = (x_2 - x_0)/N = 2/N$, so $N \geq 2/(0.002828) = 707.106$. **Take** $N = 708$.

3. Suppose that S is a natural cubic spline on $[0, 2]$ with

$$S(x) = \begin{cases} -x + 2(1-x) + a(1-x)^3 + \dfrac{2}{3}x^3, & \text{for } 0 \le x < 1, \\ b(2-x) + c(2-x)^3 + d(x-1)^3, & \text{for } 1 \le x \le 2. \end{cases}$$

Find a, b, c, and d.

Answer: Differentiate to get

$$S'(x) = \begin{cases} -3 - 3a(1-x)^2 + 2x^2, & \text{for } 0 \le x < 1, \\ -b - 3c(2-x)^2 + 3d(x-1)^2, & \text{for } 1 \le x \le 2, \end{cases}$$

and

$$S''(x) = \begin{cases} 6a(1-x) + 4x, & \text{for } 0 \le x < 1, \\ 6c(2-x) + 6d(x-1), & \text{for } 1 \le x \le 2. \end{cases}$$

Since S is a natural spline,

- $S''(0) = 0$, which gives that $a = 0$ and
- $S''(2) = 0$, which gives that $d = 0$.

To find b and c use any two of

- S is continuous at $x = 1$, which gives $b + c = -1/3$,
- S' is continuous at $x = 1$, which gives $b + 3c = 1$,
- S'' is continuous at $x = 1$, which gives $6c = 4$.

Solving any two of these gives $b = -1$ and $c = 2/3$.

Some useful formulae.

- Cauchy's theorem: If p_n is the polynomial of degree n that interpolates f at the $n+1$ points $a = x_0 < x_1 < \cdots < x_n = b$. Then, for any $x \in [a, b]$ there is a $\tau \in (a, b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!}\pi_{n+1}(x), \tag{1}$$

  where $\pi_{n+1}(x) = \prod_{i=0}^{n}(x - x_i)$ denotes the nodal polynomial.

- $\|g\|_\infty := \max_{a \le x \le b} |g(x)|$.

- If l is the linear spline interpolant to a function f on the equally spaced points $a = x_0 < x_1 \cdots < x_N = b$ with $h = x_i - x_{i-1} = (b-a)/N$, then

$$\|f - l\|_\infty \le \frac{h^2}{8}\|f''\|_\infty. \tag{2}$$